Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C, such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. □

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C. Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C, then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i/\mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \dots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

(b) Show that this is false for varieties of dimension $\geqslant 2$. In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

4. Let C be a curve of genus g, and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

(a) Show that $r = 0$ and $r \geqslant g + 1$ are always possible.

(b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

5. *Values of* e. Let C be a curve of genus $g \geqslant 1$.

(a) Show that for each $0 \leqslant e \leqslant 2g - 2$ there is a ruled suface X over C with invariant e, corresponding to an indecomposable $\mathcal{E}$.

(b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}_X(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_X(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(X, \mathcal{O}_X(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \nsubseteq H$.

(c) Now show that if $-g \leqslant e < 0$, there exists a ruled surface X over C with invariant e.

(d) For $g = 2$, show that $e \geqslant -2$ is also necessary for the existence of X.

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of th ebase curve C, and that no tow are linearly equivalent.

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \to \mathcal{F} \to 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F})/\operatorname{rank} \mathcal{F} > (\deg \mathcal{E})/\operatorname{rank} \mathcal{E}$$

Replacing $>$ by $\geqslant$ defines *semistable.*

(a) A decomposable $\mathcal{E}$ is never stable.

(b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E}.0$ (respectively, $\geqslant 0$).

(c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not seistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leqslant 2g-2$, (2) an element $\mathcal{L} \in \operatorname{Pic} C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(X, \mathcal{L}^{-1})$, deterimned up to a nonzero scalar multiple.

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geqslant 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.

10. For any $n > e \geqslant 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geqslant 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geqslant 4$, there exists a nonhyperelliptic curve which has a g^1_3.

11. Let X be a ruled surface over the curve C, defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{O}_C(\mathfrak{e}) \cong \wedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C.

(a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no points.

(b) If $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C, and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \mathfrak{e} - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.

12. Let X be a ruled surface with invariant e over an elliptic curve C, and let $\mathfrak{b}$ be a divisor on C.

(a) If $\deg \mathfrak{b} \geqslant \mathfrak{e} + 2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.

(b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geqslant e + 3$.

13. For every $e \geqslant -1$ and $n \geqslant e+3$, there i san elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d_1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.

14. Let X be a ruled surface over a curve of genus g, with invariant $e < 0$, and assume that $\operatorname{char} k = p > 0$ and $g \geqslant 2$.

(a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0.f$, then either $a = 1$, $b \geqslant 0$, or $2 \leqslant a \leqslant p-1$, $b \geqslant \frac{1}{2}ae$, or $a \geqslant p$, $b \geqslant \frac{1}{2}ae + 1 - g$.

(b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g-1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.

15. *Funny behaviour in characteristic* p. Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.

(a) Show that the action of the k-linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.

(b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.

(c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C. Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi : Y \to C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

17. (a) Let $\varphi : \mathbb{P}^1_k \to \mathbb{P}^3_k$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ. Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C, so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m.

(b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \to \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.